

Load Shedding in Bangladesh: Causes, Effects and

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Acknowledgement

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Abstract

Globally, load shedding is considered the last resort to halt the national power grid from getting overloaded when the demand for electricity surpasses the supply in the national electricity market. It prevents the possibility of the occurrence of a blackout. However, frequent and unplanned load shedding is a very concerning issue. Most nations have a national policy in place to deal with power network imbalances, with a strict set of actions that distribution network operators must follow. My encouragement behind this research work was that Bangladesh also has to adopt effective national policies to solve this problem. The research clearly showed how scarcity of natural resources causes load shedding to increase rapidly in Bangladesh. During this research work, while trying to connect with my respondents, I discovered that most people are aware of the cause behind load shedding and agree that it causes difficulties in their regular lives. The research revealed that load shedding causes a lot of problems in various sectors of the economy by hindering productiveness. People associated with this study agreed that both the government and the general people have to play their roles to reduce load shedding. The study uncovered that we need to look for alternative ways to fulfill the need for natural resources required to produce sufficient power and what steps the government can take to solve this problem.

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Introduction

The term 'load shedding' refers to the intentional stoppage of electrical energy in certain parts of the current distribution system when the demand for electricity surpasses the supply. The purpose of load shedding is to keep a power grid or power supply from becoming overloaded.

Power outages caused by load shedding can last anywhere from a few minutes to many hours. It has the potential to disrupt companies and services, diminish productivity, and irritate consumers or clients. Load shedding does more harm than good in Bangladesh. Nowadays, load shedding is happening throughout the whole country frequently and the problem is escalating day by day.

There are many reasons behind frequent power cuts. One of the main reasons is the lack of natural resources used to produce adequate electrical energy. Many of the power plants responsible for the production of electricity had to be closed because at present not only is there a scarcity of coal and fuel oil but there is also a dollar shortage in the country resulting in the inability to import resources to produce electricity. There is a lack of transparency in the energy business. Even though the government has taken initiatives to increase the supply of electricity by constructing new power plants on a rental basis, the demand has not increased by the same proportions due to various underlying reasons.

Load shedding adversely affects the lives of the general people. Students are the worst sufferers because at present the education system is highly technology-dependent. Besides, they are not getting the congenial environment required to sit to study. Thus, it brings about a lot of physical and mental health complexities. It also causes frustration among the youth since a few of them are a part of the gamer community. Furthermore, load shedding affects household sector negatively by limiting the supply of fresh water or gas. Under these kinds of circumstances,

people cannot cook and also suffers from other problems. Load shedding also affects the economy of Bangladesh negatively by decreasing productivity. The production in the mills and factories comes to a standstill and thereby adequate number of products cannot be produced timely. Still, the industries have to pay the fixed cost for capital and labor. Moreover, load shedding leads to the closure of shopping malls and other commercial institutions such as banks and offices. In this way, load shedding is affecting the overall GDP of the economy by reducing the profits earned by commercial institutions, and these industries are considered to be an essential component of the economy.

Another important sector, namely, the agricultural sector is severely affected. The production of various crops, fruits, and vegetables is being hampered since the poor farmers are struggling to operate machinery used to irrigate dry lands without electricity. According to Sheikh (2023), the director of Frozen Foods Exporters Association (BFEEA) stated that electrical energy is a necessity in every step of shrimp processing. Many experts have come forward with a lot of suggestions as to how this power crisis could be solved. For instance, the introduction of household on-grid solar systems, preventing tax breaks for the use of energy-saving appliances, changes in government policies, etc. It is high time the government took the necessary steps to mitigate this problem since it could worsen the state of the economy. If load shedding keeps on increasing at the current rate, it will be detrimental to the growth of the economy as it reduces a great deal of productivity in some of the most important sectors of the economy, thereby bringing about misery in the lives of the youth, poor people and official workers. The focus of my research will be on why load shedding is escalating in Bangladesh, what effects it might have on different sectors of the economy, and how the nation can possibly overcome this ongoing crisis.

Background

The phenomenon of load shedding is considered the last hope when the national electricity market is in a disequilibrium condition, that is, there is a shortage of electric power. Load shedding is necessary because it sustains the power grid's coherence by limiting the supply of electricity to some consumers and it helps to prevent prolonged power outages in the future which could destroy the power grid's functionality. Load shedding works as a sort of load control by rotating power outages or limiting power consumption from primary sources until demand drops and more capacity becomes available. When demand falls or new power supplies become available, providers restore electricity to the impacted areas. Load shedding is usually pre-planned and discussed with the residents of a particular area but sometimes it is imposed by the electricity suppliers and results in a sudden power cut in a specific region. This type of load shedding is known as rolling blackouts. Brownouts, another sort of involuntary load shedding, are caused by a power supplier decreasing voltage distribution during peak consumption periods to balance supply and demand. Load shedding is a very concerning issue for developing countries since it causes inconvenience in various sectors of the economies of those countries. Load shedding has risen over the past few years in Bangladesh for various reasons. At present, there are 153 power plants in Bangladesh and the nation's current power generation capacity is 23,370 MW. According to PDB data, from May 24, 2023, to May 29, 2023, 4,295 MW of power-producing capacity, or roughly 18% of the total capacity, could not be used daily due to fuel scarcity. (Mohiuddin, 2023). Plants with a minimum output of 2000 MW must be shut down for routine maintenance. The government is pressured to import gas, coal, and fuel oil because of

the country's dollar shortage. Also, the Ukraine war has added to the power shortage. The increase in the price of energy was worldwide and Bangladesh has been affected heavily. The price of LNG per MMBTU increased to \$85 in March of 2022. It dropped for a few months but then rose again to more than \$50 (Hossan, 2022).

Power plants are not receiving the necessary fuel supply as a result. Besides, there is a lack of openness in the energy business. The Office of the Comptroller and Auditor General (CAG) has found 19 charges of irregularities by Bangladesh Oil, Gas and Mineral Corporation (Petrobangla) and Bangladesh Petroleum Corporation which have resulted in a loss to the government (in fact, the people) of roughly Tk 4,697 crore. (Jamal, 2022). In comparison to overall demand, Bangladesh has more than 70% of the power-producing capacity. However, because there is less demand than capacity, the government must pay millions of crores of rupees in capacity payments every year. Many individuals feel that the government's subsidies to boost this capacity are to be blamed for the present electrical crisis. However, demand did not rise at the expected annual pace. Electrical equipment in our houses is now far more energy efficient than they were previously.

There has been an increment in the power outages in slum areas after the shutdown of the Payra Thermal Power Plant on May 25, 2023. Due to the lack of electricity, it has become hard to avail pure drinking water as the water pumps in slum areas are being destroyed due to ongoing load shedding. As a result, many of the slum residents are failing to prepare food and are falling ill due to heat. Besides, Load shedding led to the closure of many educational institutions since most of them did not have generators or IPS. Due to immense heat, many students are suffering from a lot of health complications and thus are not able to continue their studies properly. Excessive load shedding has been disrupting online classes and academic activities since

students nowadays are more dependent on electronic gadgets. Load shedding greatly hampers the economic development of Bangladesh. Apart from the industrial sector, load shedding is causing small-scale factories, poultry farms, dairy farms, fish processing industries, and textile industries to be closed. As a result, ordinary farmers other proprietors, and weavers are leading miserable lives. The cost of living has been increased due to load shedding since the owners of generators have to pay excessive bills for running generators. The Power Development Board has been facing great losses due to the increased price of fuel. According to Halder et al. (2022), garment factories had to incur a loss of \$50 million since they could produce less using generators. In addition, cultivable lands are becoming dry because of natural disasters like drought in the Barendra region but the farmers are unable to irrigate those lands frequently since diesel has become very costly. Besides, the water pumps and deep tube wells are likely to shut down because of frequent load shedding. As a result, there is a great risk of destruction of the production quality of various crops and fruits such as Ayush and Aman paddies. It is also causing great losses to the export-oriented fisheries sector in Khulna district of Bangladesh. The organization named West Zone Power Distribution Company Limited which is responsible for power supply in Khulna has been unable to meet the excess demand for electricity. As a result, it is not possible to preserve perishable products like shrimp at accurate temperatures.

Many experts have come forward with possible solutions for load shedding. The power crisis can be solved by introducing household on-grid solar systems because they are capable of producing more than 1kW to 10kW compared to the traditional DC off-grid systems (Mahmud, 2022). “In order to get out of dependence on energy imports, the government must undertake transparent and corruption-free implementation of the National Solar Energy Roadmap 2021-41” (Taiyeb, 2023, para. 24). To overcome the problem of load shedding in Bangladesh, awareness is

necessary among the general people so that they can be frugal when it comes to using electricity. Such as turning off electrical switches and appliances after the completion of work, sleeping without lights on, etc. A few initiatives can be taken by the government to reduce load shedding such as keeping educational institutions closed for 15-30 days, reducing office hours, etc. The government can adopt some short-, medium- and long-term policies to mitigate load shedding such as tax breaks provided for the use of energy-saving AC, washing machines, ovens, refrigerators, and elevators, and there should be a clear work plan in place to preserve energy by preventing the theft and corruption that is responsible for structural system losses. Above all, awareness among the general people and the government is required to solve this issue.

Research Questions

In my research, I intend to look at the causes, consequences, and solutions to load shedding in Bangladesh. My study will answer the following research questions:

- 1) What exactly is load shedding?
- 2) What are the reasons for power outages in Bangladesh?
- 3) How does load shedding influence youngsters and the wider public?
- 4) How does load shedding affect the industrial, agricultural, and fisheries sectors?
- 5) What are some potential solutions to load shedding in Bangladesh?

Hypothesis

The negative aspects of load shedding outweigh the positive aspects in the context of a developing country like Bangladesh. From my research, I expect to find that the scarcity of natural gas, coal, and oil as well as improper maintenance, and lack of openness in the energy

sector limit the power supply. I presume that the youth is facing a lot of difficulties due to load shedding as they are dependent on technology for both educational and recreational purposes. I think that load shedding will take a toll on people irrespective of their profession and age. It is expected that load shedding will impede economic development by negatively affecting the industrial, fisheries and agricultural sectors as productiveness will be reduced. I also expect that there will be a way out of this problem by making a few changes in the existing policies and finding alternatives to produce energy.

Research Methodology

For my primary research, I have administered an online sample survey using “Google Forms”. A vast majority of the respondents were students of various educational institutions and other professionals like jobholders, teachers, homemakers, businesspersons and retired. The aim of my survey is to learn about the experiences people had whenever load shedding occurred and find out what are their thoughts on the current energy crisis in the country and their recommendations to overcome this problem. The majority of my surveys contain multiple-choice questions on my study subject.

For my secondary research, I used various online articles, journals, newspapers and websites on my research topic which assisted me to become more knowledgeable about the true situation.

Data Presentation and Analysis

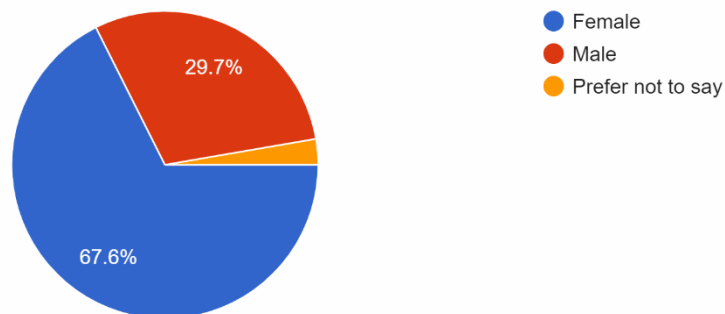
In order to collect the data, I conducted an online sample survey of 37 individuals using “Google Forms,” as it was impossible to conduct an offline survey because of time constraints. The intention behind conducting this survey was to know people’s points of view on the causes, effects, and solutions of load shedding in Bangladesh.

The survey questions are given below. The first three questions are demographic questions; the rest are my real inquiries.

Question 1: Gender: -

Gender:

37 responses



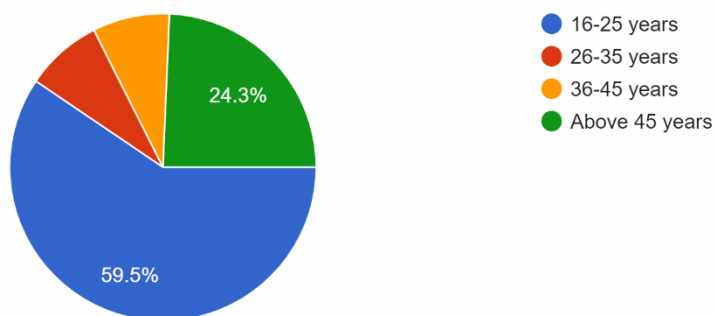
The pie chart above illustrates the gender preference for this survey.

As load shedding is thought to be impacting every people irrespective of their genders, we preferred every gender. Among 37 respondents, 25 (67.6%) were females, 11 (29.7%) were males and 1 (2.7%) preferred not to say their genders.

Question 2: Age: -

Age:

37 responses



The pie chart above shows the age group of respondents.

Since it was an online survey, we could avail responses from people who were active online.

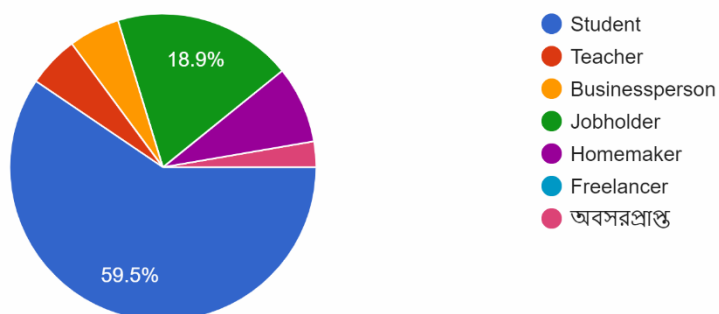
Among 37 respondents, 22 (59.5%) were young adults of 16-25 years who were the most active.

9 (24.3%) were adults above 45 years who were the second most involved. 3 (8.1%) were individuals of the age group 26-35 years and the other 3 (8.1%) were 36-45 years old and were the least involved online.

Question 3: What is your profession?

What is your profession?

37 responses

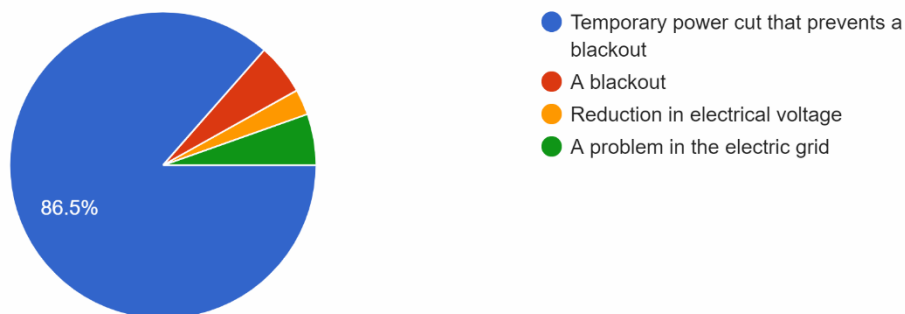


The pie chart above shows the professions of my respondents.

Among 37 respondents, 22 (59.5%) were students, 7 (18.9%) were jobholders, 3 (8.1%) were homemakers, 2 (5.4%) were businesspersons, 2 (5.4%) were teachers and 1 (2.7%) is retired. The motive behind collecting views from people of various professions was to get more varied opinions about the causes, effects, and remedies of load shedding in Bangladesh from different perspectives.

Question 4: What do you understand by load shedding?

What do you understand by load shedding?
37 responses



Through this question, I wanted to find out whether people are able to differentiate between blackout, brownout, and load shedding or not.

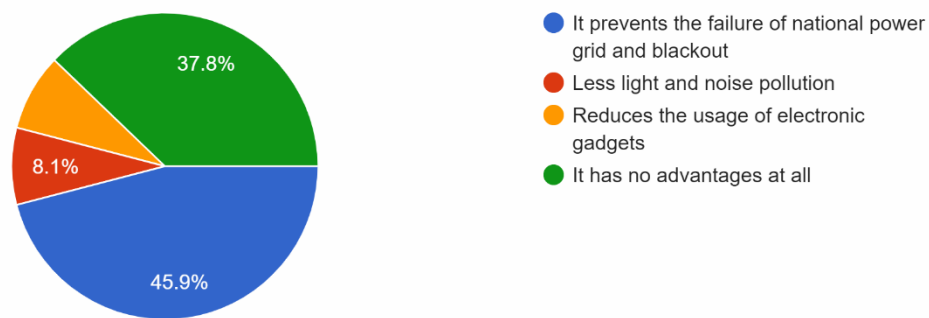
The above pie chart shows that the majority of the respondents, who are 86.5% of the total could identify the true meaning of load shedding, maybe because load shedding occurs every now and then in Bangladesh. The chart also shows that 5.4% of the respondents mistook load shedding to be a blackout, maybe because people in some regions of the country have to experience power outages lasting from 6-7 hours. 2.7% of the respondents presumed load shedding to be a

reduction in the electrical voltage, that is, a brownout. The rest 5.4% thought that load shedding is a problem in the power grid, maybe because they are aware of the improper maintenance of power plants.

Question 5: What are your thoughts on the advantages of load shedding?

What are your thoughts on the advantages of load shedding?

37 responses



I asked my respondents this question to know whether they were aware of the necessity of load shedding.

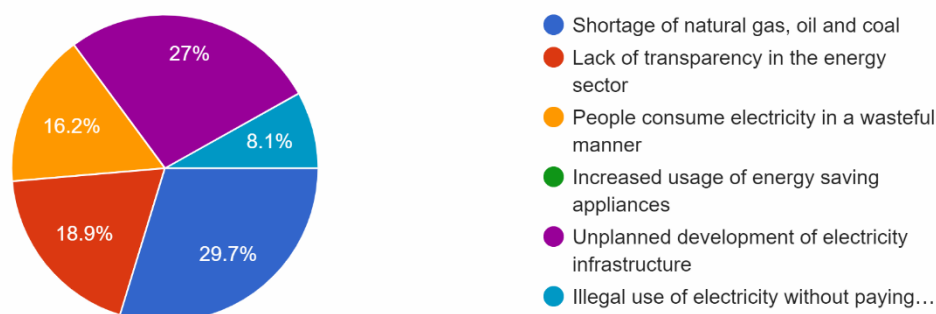
From the above pie chart, we can observe that the maximum number of people, who are 45.9% of the total respondents, know the actual logic behind the occurrence of load shedding, that is, it prevents the failure of the national power grid. It may be because they know that load shedding is needed to preserve the integrity of the power grid by load management. However, 37.8% of the respondents thought that load shedding has no advantages at all, maybe because they are residents from the regions where load shedding occurs routinely and they have to suffer greatly because of that. 8.1% thought that load shedding causes less light and noise pollution, maybe because they have elderly people and children in their houses who are very sensitive to loud noises and bright light. The rest 8.1% thought that it reduces the usage of electronic gadgets, it

could be because most electrical gadgets cannot run without a constant power supply or net connections which could provide scopes to engage in offline physical activities.

Question 6: What is the main cause of increasing load shedding in Bangladesh?

What is the main cause of increasing load shedding in Bangladesh?

37 responses



The intention behind asking this question was to find out what could be the main reason among many speculated reasons for increasing load shedding.

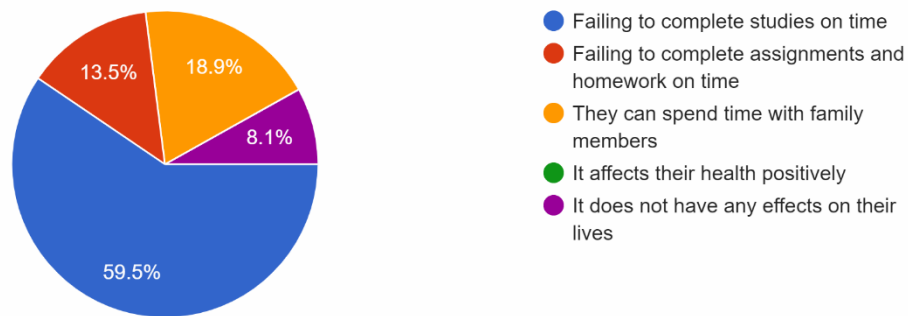
From the pie chart, it is seen that a maximum number of people, who are 29.7% of the respondents thought that the shortage of natural gas, coal, and oil is the main reason, maybe because they are aware of the fact that Bangladesh is currently ongoing dollar shortage leading to the inability of importing natural resources which are the main ingredients of producing power. 27% of the respondents thought that unplanned development of electricity infrastructure is the main reason, maybe because they know that many of the established power plants are being closed down or sitting idle because of lack of resources and maintenance. 18.9% of the respondents thought that lack of transparency in the energy sector is the main reason, maybe because the officials involved in those sectors are corrupt or not doing much to solve this problem. 16.2% of the respondents thought that people consume electricity in a wasteful manner

which could lead to a loss of a huge amount of power. 8.1% thought illegal use of electricity without paying for it to be the main reason and 0% thought increased use of energy-saving appliances could contribute to increased load shedding.

Question 7: What is the impact of load shedding on youth?

What is the impact of load shedding on youth?

37 responses



My intention behind asking this question to my respondents was to know whether load shedding affects young adults and adolescents positively or negatively.

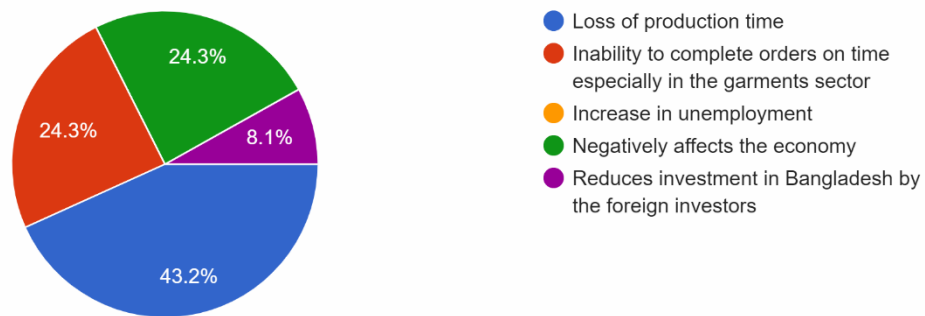
From the pie chart, it is evident that the maximum number of respondents, who are 59.5% of the total thought that failing to complete studies on time is the crucial problem faced by the youth, which may be because of the absence of generators or IPS in houses which could create an unfavorable condition to sit to study. 18.9% thought that the youth can spend time with their family members, maybe because load shedding could prevent young adults from getting too involved in online activities using electricity run devices, thereby providing them with opportunities to spend time with loved ones. 13.5% thought that the youth is failing to complete assignments and homework on time, maybe because many of the assignments need to be submitted online or to be completed using laptops which is not possible without a power supply.

8.1% thought that it does not have any effects on their lives and 0% thought it affects their health positively.

Question 8: How does load shedding affect the industrial sector?

How does load shedding affect the industrial sector?

37 responses



I intended this question to know what is the major problem faced by the industrial sector due to frequent load shedding.

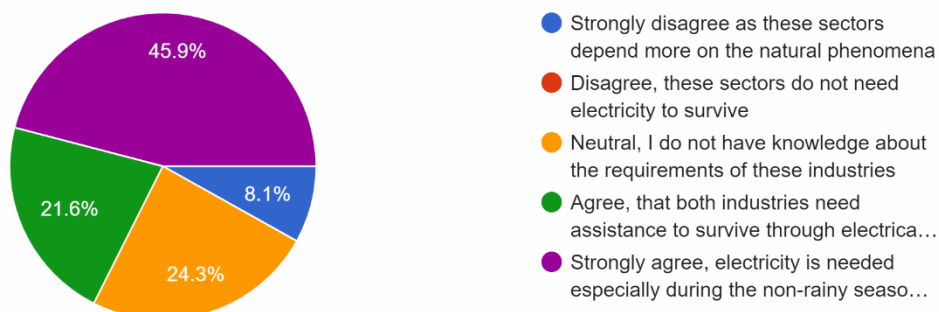
From the pie chart, the maximum number of respondents, who are 43.2% of the total thought loss of production time is the most negative effect, maybe because during load shedding the machinery in the factories comes to a standstill and workers cannot complete production of goods on time. 24.3% of the respondents thought that the inability to complete orders on time, especially in the garments sector is the main problem, it could be because garment workers use sewing machines run by electricity to make garments rather than hand sewing which is stopped during load shedding leading to decreased productivity. 24.3% thought that it negatively affects the economy, maybe because many industries fail to produce an adequate amount of goods using generators which raises their production cost and reduces profits leading to an overall decline in GDP. 8.1% thought that it reduces investment by foreign investors in Bangladesh, maybe

because the industrial sector becomes less productive and fails to complete orders on time which discourages foreign investors. 0% thought that this problem increases unemployment.

Question 9: How much do you agree or disagree with the statement that load shedding affects the agriculture and fisheries sector of Bangladesh adversely?

How much do you agree or disagree with the statement that load shedding affects the agriculture and fisheries sector of Bangladesh adversely?

37 responses



I asked this question to my respondents to find out if they were aware that load shedding could have a negative impact on the agriculture and fisheries sector.

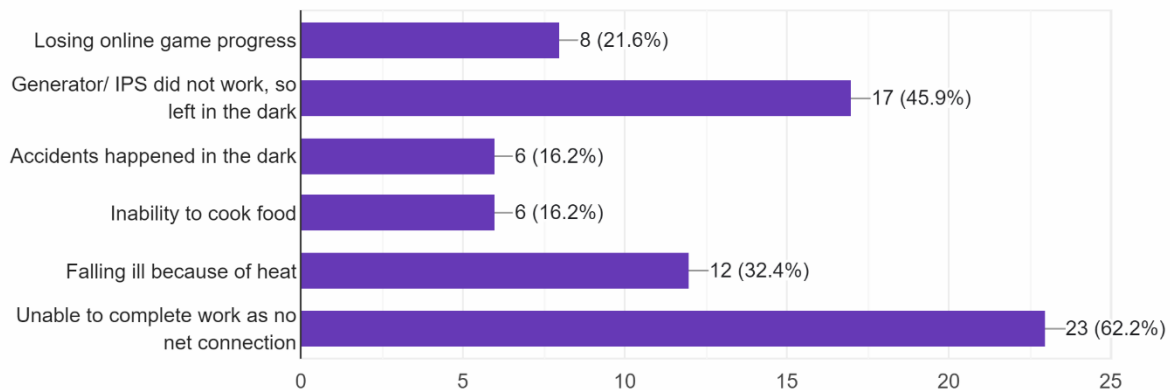
From the pie chart, it can be observed that the maximum number of respondents, who are 45.9% of the total strongly agree with the statement, maybe because electricity is needed to water the fields, especially during the non-rainy seasons and fisheries are now using hatcheries for which electricity is needed. 24.3% neither agree nor disagree, maybe because they do not have any knowledge about the requirements of these sectors. 21.8% of the respondents agree that both industries need assistance to survive through electrical supplies. However, 8.1% strongly disagree that load shedding negatively impacts the agriculture and fisheries sector because they

presume these sectors to be dependent mostly on natural phenomena and 0% disagree that these sectors do not need electricity to survive.

Question 10: What kind of difficulty have you faced the most during load shedding?

What kind of difficulty have you faced the most during load shedding?

37 responses



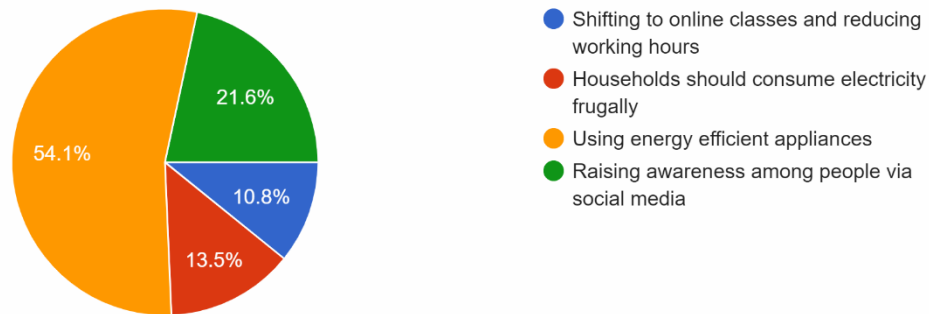
The bar graph above illustrates the problems faced by the general people during load shedding.

From the graph, it is evident that the maximum respondents, who are 62.2% of the total were unable to complete work as there was no net connection, which may be because they could not utilize the resources available online and incorporate them into their work. 32.4% fell ill because of heat, perhaps they did not have a generator or personal IPS so the ceiling fan could not operate. 45.9% reported that the generator/ IPS did not work, so left in the dark. This might be because the oil used to operate the generator ran out or the IPS was damaged because of recurring electricity. 21.6% lost online gaming progress, maybe because their gaming PC or devices needing uninterrupted power supply suddenly shut down due to load shedding. 16.2% were unable to cook food and the rest 16.2% reported that accidents happened in the dark.

Question 11: What could be done to mitigate load shedding?

What could be done to mitigate load shedding?

37 responses



My intention behind asking this question was to know what the general public thought to be the remedy to this problem.

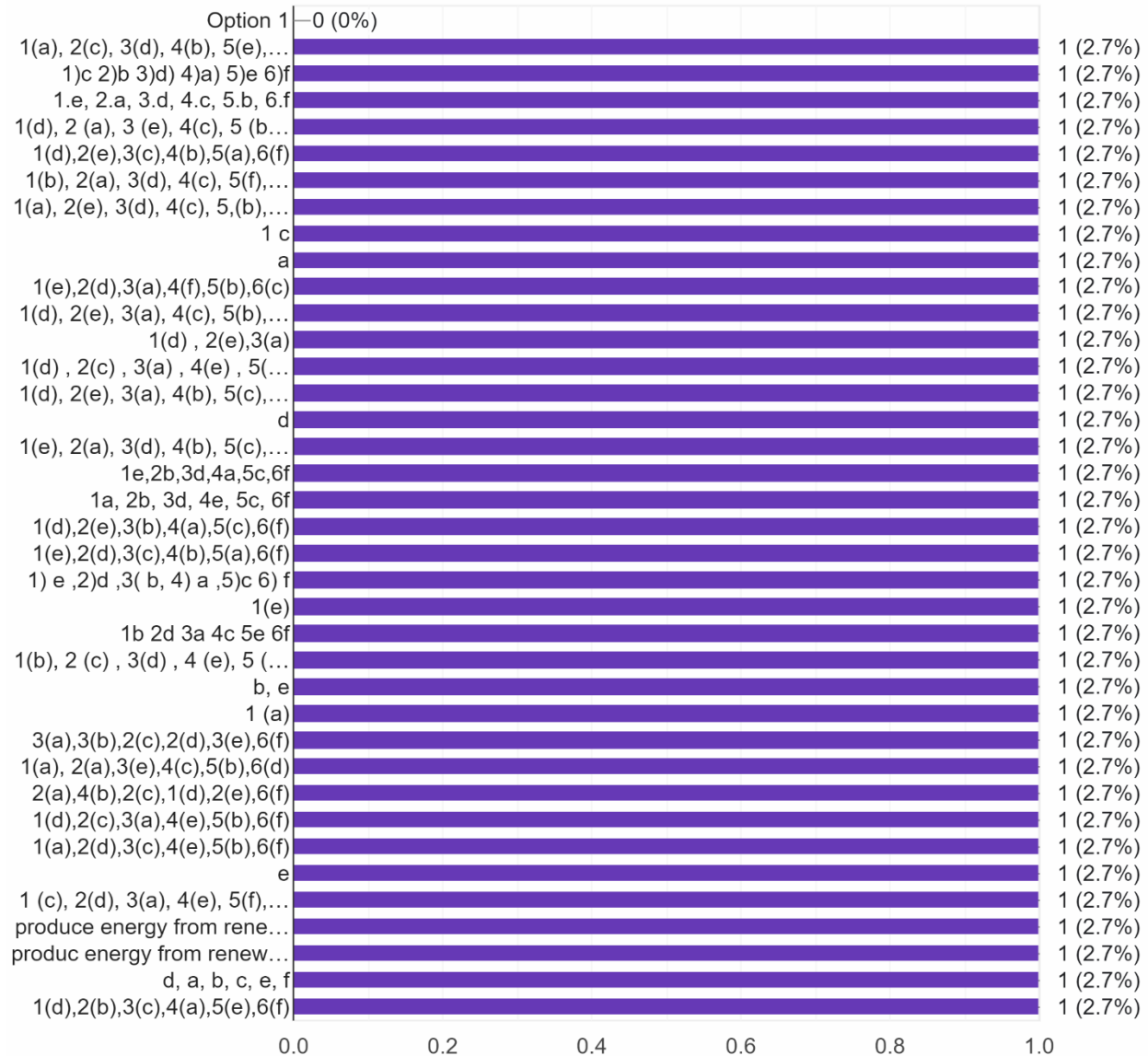
From the pie chart, the maximum respondents, who are 54.1% of the total thought using energy-efficient appliances could mitigate load shedding, maybe because energy-saving appliances could not only reduce electricity bills but also could prevent excess use of electricity. 21.6% thought that raising awareness among people via social media could be a viable solution, maybe because some people who are not aware of this crisis might come forward to assist the government in solving this problem. 13.5% thought that households should consume electricity frugally, maybe because some people are not careful when it comes to using electricity and leave lights and fans on even when they do not need them. The rest 10.8% thought shifting to online classes and reducing working hours could be a possible solution, maybe because it reduces overall power consumption by reducing the use of electricity appliances in a large number of institutions.

Question 12: What steps can the government take to solve the problem of frequent load shedding? Number your choice from 1-6 with 1 being the most effective and 6 being the least effective.

- a) Introducing household on-grid solar systems
- b) Exploring and digging gas fields available in the country
- c) Decreasing dependency on the import of natural gas
- d) Produce energy from renewable sources
- e) Check illegal connections of electricity supply
- f) Increase cost of electricity per unit

What steps can the government take to solve the problem of frequent load shedding? Number your choice from 1-6 with 1 being the most effective and 6...he least effective. [For example, 1(e), 2(b)...6(a)]

37 responses



The motive behind asking this question to my respondents was to know which of the above-mentioned steps they thought the government could take to reduce load shedding. We got 37 different answers from 37 different respondents.

From the graph, it can be seen that 23 out of 37 respondents thought increasing the cost of electricity per unit to be the least effective measure, maybe because it could increase theft and illegal connection of electricity leading to even more load shedding. 13 out of 37 respondents thought that producing energy from renewable resources is the most viable solution, maybe because they are inexhaustible and it might be possible to use these innumerable times to produce energy. 7 out of 37 respondents thought that introducing household on-grid solar systems could solve this problem, it might be because both the households and the national electricity market could benefit from it and 6 out of 37 respondents thought that checking illegal connections of electricity could reduce load shedding, maybe because a lot of power could be conserved and used for beneficial purposes by punishing those involved with this kind of activity. 4 out of 37 respondents thought exploring and digging gas fields available in the country to be the most effective solution, perhaps this could fulfill the shortage of natural resources and only 3 respondents thought that Bangladesh should decrease dependency on the import of natural gas.

Summary of Research Findings

The most interesting finding of this research work is that the shortage of natural gas, coal and oil is the root cause of increasing load shedding in Bangladesh. Most people are aware of the fact that our country is facing difficulties in importing natural resources from international markets as the prices of these resources have increased because of the Russia-Ukraine war. The power

plants cannot generate power because of the scarcity of these resources. Besides, the unplanned electricity infrastructure development is to blame for this problem.

Load shedding has increased in Bangladesh over the past few decades, so a vast majority of the people are aware of this phenomenon. Some people think that the general public is to blame for the increase in load shedding since many consume electricity in a wasteful manner. Besides, some think that the government is not checking illegal connections of electricity properly and imposing punishments.

Load shedding is supposed to be beneficial for maintaining the integrity of the national power grid but it has become a serious problem now. One of the main problems caused by load shedding is that it diminishes productivity in every sector of the economy, for instance, the industrial sector, the educational sector, the agriculture and fisheries sector. It obstructs people from each of these sectors from completing their due work on time. Both the government and the people should come forward to solve this problem.

The findings of my research have almost matched my expectations and hypothesis in a substantial way. The research completely supported the fact that load shedding takes a toll on people irrespective of their profession and age by preventing them from finishing their tasks or by causing other problems in their day-to-day lives. However, it is possible to solve this problem by thinking of ways to generate power from renewable sources and by taking other measures.

The research definitely has some limitations. The reason for the occurrence of these limitations is that the time given to finish the research study was finite. Hence, it was not possible for me to take personal interviews with people and visit different sites.

Recommendation

From the research, it is evident that a vast majority of the people are aware of the actual logic behind load shedding and they suggested that the government should look for alternatives such as producing electricity from renewable sources rather than relying heavily on imports. Using renewable energy sources can bring stability in the price of electricity which can prevent people from adopting illegal means to consume electricity. Besides, it will create economic development by decreasing reliance on imports. In addition, the introduction of household on-grid solar systems can reduce the load on the national grid and provide energy to the industrial sector as well. The general people and the government must come forward to solve this problem. I would recommend people use electricity-efficient appliances in their houses and raise awareness via social media platforms to consume electricity frugally.

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